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Paleomagnetic Age of the Central Missouri Barite Deposits and Its Genetic Implications

D. T. A. SYMONS

Department of Geology, University of Windsor, Windsor, Ontario, Canada N9B 3P4

AND D. F. SANGSTER

Geological Survey of Canada, 601 Booth Street, Ottawa, Ontario, Canada K1A 0E8

Abstract

In the Central Missouri barite district, barite samples were taken from four barite deposits and carbonate samples from the Lower Ordovician Jefferson City and Gasconade Dolomites in two road cuts and from the Mississippian Burlington Limestone in a quarry. The samples were tested paleomagnetically using detailed alternating field and thermal demagnetization and saturation isothermal remanent magnetization methods. An A magnetization direction of declination = 153.5°, inclination = 17.6° (N = 35, k = 17, α_{95} = 4.8°) is carried by pseudosingle domain magnetite in the barite-rich zones of three open-pit mines as an ore-synchronous component, in the Jefferson City Dolomite as a secondary component, and in collapse breccia blocks of probable post-Middle Devonian age in the dolomite to give a negative "conglomerate" test. The A paleopole at 121.0°E, 36.6°N (dp = 2.6°, dm = 4.9°) falls within the Kiaman reversed magnetic polarity interval on the Late Pennsylvanian to early Early Permian part of the apparent polar wander path for North America. A B magnetization direction of declination = 36.8°, inclination = 77.8° (N = 21, k = 13, α_{95} = 92°) is found in the other units. It is a hybrid magnetization dominated by hematite produced by weathering and by viscous remanence in magnetite, both of Quaternary origin. The Kiaman magnetization in the barite ores and host rocks of the Central Missouri district agrees with known Kiaman magnetizations in the lead-zinc ores and host rocks of the adjacent Southeastern Missouri and Northern Arkansas districts. This agreement provides strong support for the Ouachita tectonic uplift model of genesis for the origin of the Mississippi Valley-type ore deposits in the midcontinental region of the United States.

Introduction

IN THE Central Missouri district, many small high-grade barite deposits occur as open-space fillings in solution collapse structures in carbonate rocks. Although these deposits are thought to be genetically related to the Mississippi Valley-type barite and lead-zinc deposits of the Southeastern Missouri, Northern Arkansas, and Tri-State districts (Leach and Rowan, 1986), their relationship is difficult to establish because of uncertainties in the age(s) of mineralization (Sangster, 1986). The objective of this study is to use the paleomagnetic method to determine the age of the barite mineralization. In Mississippi Valley-type ores and host rocks, chemical remanent magnetization is acquired by chemical precipitation during initial carbonate sedimentation or subsequent chemical alteration of the ore and ganque minerals and is subsequently measured.

Geology

The geology of the Central Missouri barite district has been summarized by Leach (1980). The barite deposits occur in a structural high along the northwestern flank of the Ozark uplift, centrally located between the Forest City, Illinois, and Arkoma basins (Fig. 1a). The deposits are found throughout a gently dipping sedimentary sequence that ranges in age from Late Cambrian to Early Pennsylvanian and is composed mainly of carbonates with coals, shales, and sandstones prominent in the Pennsylvanian part of the succession. Most of the barite deposits, however, are hosted by the Lower Ordovician Jefferson City and Gasconade Dolomite Formations of the Canadian

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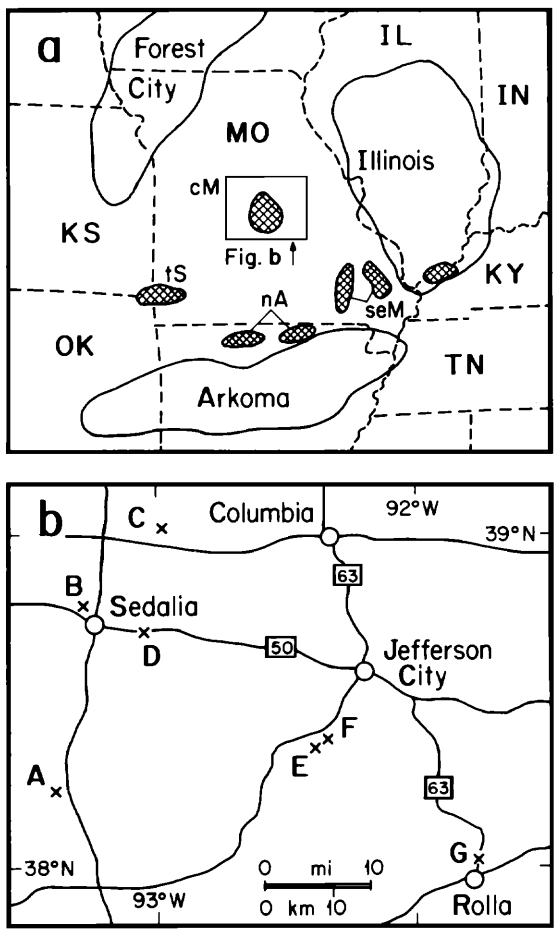


FIG. 1. Location maps. (a). The Missouri area with Mississippi Valley-type ore districts shown. cM = the Central Missouri district, seM = the Southeastern Missouri district, nA = the Northern Arkansas district, and tS = the Tri-State district. (b). The Central Missouri district with sampling locations. A = Blackman pit, B = Manafee quarry, C = Widel pit, D = highway 50 road cut, E = Goller pit, F = Stenbergen pit, and G = highway 63 road cut.

Series and are located preferentially in solution collapse structures along the crests of gentle anticlinal flexures. These structures include solution fissure and breccia zones, solution channels, filled sinks, and solution circle deposits. The barite both replaces and cements the original fragments and penetrates both fragments and wall rocks as very narrow, irregular veins.

The paragenetic sequence for the deposits is simple: pyrite-marcasite first, followed by galena, sphalerite, barite, and calcite with chalcopyrite occurring throughout the sequence (Leach, 1980). The fine-grained massive white barite is by far the dominant economic mineral. Fluid inclusion studies in central Missouri and the surrounding area (Leach, 1979, 1980; Coveney et al., 1987) indicate that sphalerite precipitated from brines at 80° to 110°C with >22 wt percent salts whereas barite fluid inclusions suggest precipitation from brines at <50°C with 4 to 10 wt percent salts. The gaps in both temperature and salinity suggest that the lead-zinc and barite represent two distinct events, perhaps unrelated. The lead-zinc fluid inclusion temperatures are anomalously high considering that burial depths in the area probably did not exceed 1 km (Howe, 1961). One possible explanation is that of upward migration of hot brines from a deeper basin. The time controls are minimal. The barite deposits could have been formed in one event postdating the Early Pennsylvanian or in multiple events starting as early as the Late Cambrian.

Sampling

Sixty-seven large oriented block samples were collected from seven locations (Fig. 1b). At the Blackman (blocks 1–12), Goller (13–18), and Stenbergen (19–21) barite pits, each block was taken from a different barite-rich zone on the pit walls, well below the original suboutcrop level. The Widel blocks (17–37) were collected from surface outcrops surrounding the present filled-in pits. The Blackman, Goller, and Stenbergen deposits are hosted by the Jefferson City Dolomite and the Widel deposit by the Mississippian Burlington Limestone of the Osagean Series. Three unmineralized stratigraphic sections were also sampled (Fig. 1b, locations B, D, and G). In these, the blocks were collected at 1-m intervals in ascending stratigraphic order. The road cut (22–26) for state highway 50 is in silicified Gasconade Dolomite at the base of the Ordovician; the Manafee quarry (28–49) is in the Burlington Limestone; and the highway 63 road cut (50–58) is in the Jefferson City Dolomite. The strata at all these locations are within about 2° of horizontal so tilt corrections were not applied to the remanence directions. Also in the long highway 63 road cut is a well-exposed collapse breccia with fragments commonly exceeding 1 m in diameter, and here each block (59–67) was collected from a fragment to provide a conglomerate test.

Experimental Methods

From three to six specimens—most commonly four—were drilled from each block (Table 1) for analysis to give a total of 277 specimens. The specimens were stored for two weeks inside a steel shielded room with an ambient magnetic field of ≤100 nT to allow their viscous remanent magnetization components to decay. All subsequent measurements were done inside this room. Remanence measurements were made with an automated two-axis CTF Systems Inc. DRM-420 cryogenic magnetometer with a sensitivity of $<4 \times 10^{-6}$ A/m. After measuring the natural remanent magnetization of all specimens, one specimen from each block was either alternating field demagnetized in 11 steps up to 160 mT using a Sapphire

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TABLE 1. Block Mean Remanence Data

B	N, n	Dec	Inc	k	α_{95}	B	N, n	Dec	Inc	k	α_{95}
1	4, 1	168	19	28	18	35	3	32	87	47	18
2	5, 1	155	32	12	23	36	3	27	60	26	25
3	4	152	14	157	7	37	4	170	68	23	20
4	5	154	26	58	10	38	4	288	69	91	10
5	4	138	−3	71	11	39	4	338	16	23	20
6	4	156	18	173	7	40	3	14	59	103	12
7	4	137	22	222	6	41	4	300	19	70	11
8	4	145	13	559	4	42	4	281	76	244	6
9	4	147	29	149	8	43	4	101	64	26	18
10	5	135	34	134	7	44	4	271	16	117	9
11	4	143	0	620	4	45	3, 1	277	−2	69	15
12	4	149	30	605	4	46	3, 1	302	23	21	27
13	3	148	35	43	13	47	4	121	65	29	17
14	4	141	32	80	10	48	3	75	74	63	12
15	3	156	24	361	6	49	4	176	71	37	15
16	3	160	−1	471	4	50	4	165	29	170	7
17	4	169	2	1,791	2	51	4	163	25	68	11
18	4	158	−5	258	6	52	4	157	26	286	5
19	4	157	5	146	8	53	4	153	16	952	3
20	4	155	15	95	9	54	3	158	−7	197	9
21	4	156	10	507	4	55	4	155	34	343	5
22	4	320	69	80	14	56	4	153	22	49	13
23	4, 1	171	14	67	11	57	4	186	32	139	8
24	3	67	45	33	22	58	4	191	11	41	15
25	3	147	81	26	24	59	4	211	41	88	10
27	4	8	84	191	7	61	4	156	7	311	5
28	4	28	66	245	6	62	3, 1	135	4	459	6
29	5	76	82	28	15	63	4	160	31	511	4
30	4	56	74	79	10	64	4	155	42	202	6
31	4	29	57	200	7	65	4	163	5	118	8
32	4	12	37	51	13	66	3, 1	153	76	356	7
33	5, 1	340	63	100	9	67	3	156	9	203	9
34	3	27	70	14	33						

Notes: *B* is the block identification with blocks 26 and 60 omitted because they retain no coherent remanence; N and n are the number of accepted and rejected specimen vectors, respectively; Dec, Inc, and α_{95} are the declination, inclination, and cone of 95 percent confidence in degrees, respectively, and k is the precision parameter of the mean vector (Fisher, 1953)

Instruments SI-4 unit or thermally demagnetized in 10 steps up to 600°C using a Schonsted TSD-1 unit. As will be shown, the response of the natural remanence to both alternating field and thermal step demagnetization proved to be both similar and simple, so that all remaining specimens were alternating field step demagnetized in 20 mT increments to 100 mT. The remanent magnetization direction for each specimen was determined using principal component analysis (Kirschvink, 1980) with maximum angular deviation values of less than 7°. Mean directions were calculated using tiered Fisher (1953) statistics (Table 1). Thirteen specimens from various units with a variety of demagnetization behaviors were tested by saturation isothermal remanent magnetization analysis. Using a Sapphire Instruments SI-6 pulse magnetizer, each specimen was magnetized in nine steps up to 0.9 T, followed by alternating field demagnetization in five steps up to 50 mT. In addition, specimens cored from unoriented single crystals of white barite, blue barite, and galena and from galena powder in plaster of paris and a cluster of sphalerite crystals were tested by saturation isothermal remanent magnetism. Finally magnetic mineral extracts were separated (Pan et al., 1990) from six representative samples to examine their morphology and chemical composition using scanning electron microscopy with energy dispersive analysis.

Results

The natural remanence intensities are in the 5×10^{-5} to 5×10^{-4} A/m range for most specimens. A few of the baritic specimens have intensities up to about 1×10^{-3} A/m and a few of the carbonate specimens have intensities down to about 1×10^{-5} A/m. The natural remanent magnetic directions are scattered but show a clear bias into two populations—one is directed moderately down to the south-southwest and the other steeply down to the north near the present earth's magnetic field direction.

On alternating field step demagnetization, an initial viscous remanent magnetism component is removed

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by about 20 mT in nearly all specimens, and thereafter, their magnetization undergoes near-linear decay to the origin of an orthogonal vector plot (Zijderveld, 1967; Figs. 2 and 3) to define two populations of directions. The first, or A component, is directed to the south-southeast with horizontal to shallow down inclinations (Fig. 2); the second, or B component, is directed steeply downward with a northerly bias (Fig. 3). The same generalization applies for thermal step demagnetization of most specimens. An initial viscous remanence component is removed by the 200° or 300°C step, above which a single component emerges with either the A or B component direction (Figs. 4 and 5). After demagnetization at either 140 mT or 600°C, the remanence intensities are typically reduced to either the mid 10^{-6} A/m range or to about 2 percent of their natural remanent magnetism intensities. The characteristic magnetization directions for about three-quarters of the alternating field demagnetized specimens are defined by data from five or more steps covering the 20 to 100 mT coercivity range with maximum angular deviation values of <<5°.

The A component

All of the baritic zone blocks from the Blackman, Goller, and Stenbergen pits retain the A magnetization direction (Table 1, Fig. 6a). Their natural remanence is reduced to less than 10 percent by alternating field demagnetization with distributed coercivities to 120 mT, indicating that magnetite is the main remanence carrier (Fig. 2, blocks 8, 17, and 21). Thermal demagnetization typically shows distributed blocking temperatures up to about 560°C, again denoting magnetite (Fig. 4, blocks 9, 11, and 20). The saturation isothermal remanence curves all show a rapid initial remanence acquisition, characteristic of magnetite, either on its own (Fig. 7a, block 15) or with trace to minor contributions of a high coercivity phase that is characteristic of hematite (Fig. 7a, blocks 4, 8, and 21).

Demagnetization of the saturation isothermal remanence shows that the magnetite carriers are mostly

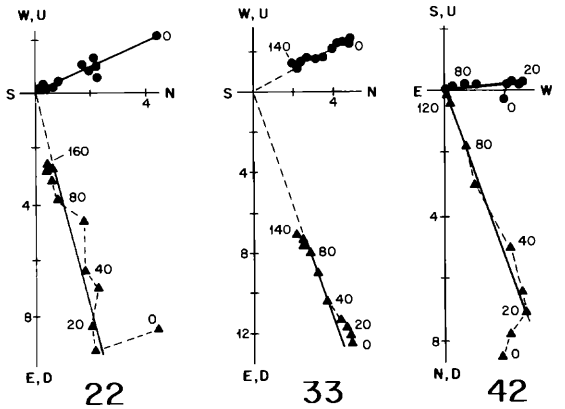


FIG. 3. Orthogonal alternating field demagnetization diagrams showing example specimens carrying the B component. Conventions are as in Figure 2.

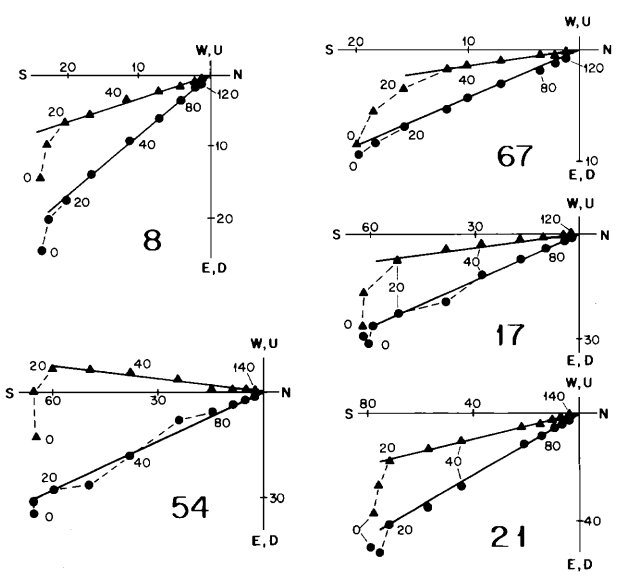


FIG. 2. Orthogonal alternating field demagnetization diagrams showing example specimens carrying the A component. The axes are north (N), east (E), south (S), west (W), up (U) and down (D). The axial units are in 10^{-5} A/m. Circles and triangles denote projections onto the horizontal and vertical planes, respectively. The alternating field step intensities are in milliTeslas. The specimen comes from the block designated by the large number.

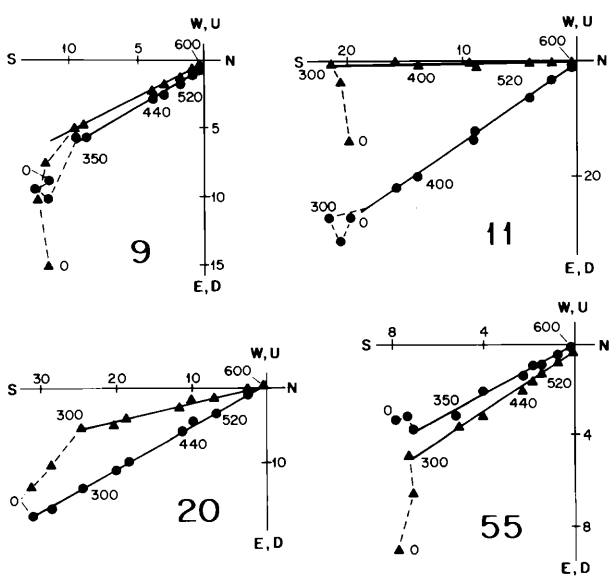


FIG. 4. Orthogonal thermal demagnetization diagrams showing example specimens carrying the A component. Conventions are as in Figure 2 except that the step temperatures are in degrees Celsius.

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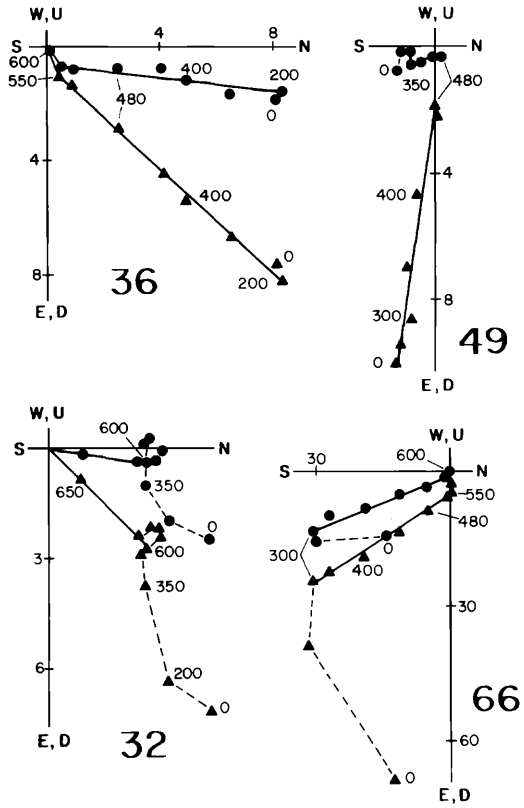


FIG. 5. Orthogonal thermal demagnetization diagrams showing example specimens carrying the B component or combined A plus B components. Conventions are as in Figure 4.

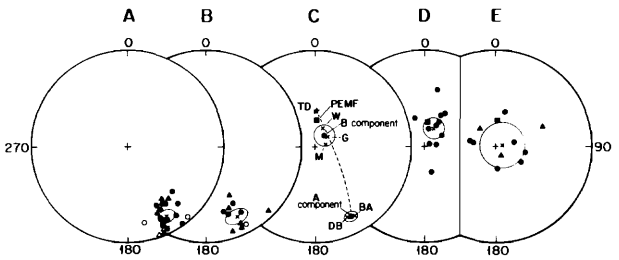


FIG. 6. Equal-area stereograms (down (up) vectors shown as solid (open) symbols). A. The A component from the Blackman (circles), Goller (triangles), and Stenbergen (squares) barite zones with their mean (×) and cone of 95 percent confidence. B. The A component in the highway 63 road cut from the dolomite section (circles) and breccia blocks (triangles) with their mean (×). C. The overall A component mean enclosing the means for the barite zones (BA) and the highway 63 dolomite and breccia (DB), and overall B component mean enclosing the means for the Widel barite zones (W), the highway 50 dolomite (G) and the Manafee quarry (M) in relation to the present earth's magnetic field direction (square, PEMF) and the earth's time-averaged dipole direction (TD) with the great-circle (dashed line) between the A component and TD directions. D. The B component from the barite zones on the Widel property (circles) with their mean (×) and the present earth's magnetic field direction (square). E. The B component in the highway 50 road cut of the silicified Gasconade Dolomite (triangles) and in the Manafee quarry section of Mississippian Burlington Limestone (circles) with the latter's mean (×).

in the single domain to pseudosingle domain size range (Fig. 7b). The high coercivities, high blocking temperatures, and saturation isothermal remanent magnetism properties of the magnetite remanence are consistent with its being a chemical remanence carried by single domain diagenetic magnetite (Jackson, 1990) that was formed at the <120°C temperatures indicated by fluid inclusion data (Leach, 1980; Coveney et al., 1987). A postemplacement thermal event to reset this A magnetization would have to reach temperatures of >350°C for 100 m.y. in order to reset the higher coercivity and blocking temperature domains (Pullaiah et al., 1975; Walton, 1980). There is no other evidence for such an event. Indeed, the conodont alteration index value for the area of 1.5 (D. Sangster, unpub. data) rules out the possibility of such a thermoviscous remanence being produced in these rocks (McCabe and Elmore, 1989). The only ferruginous crystals in the magnetic extracts from specimens from the Blackman and Goller pits are small (<3 μm) blocky crystals (Fig. 8a) that give only iron peaks on EDA, a combination that is diagnostic of magnetite.

The mean magnetization directions for the barite zones in the adjacent Goller (blocks 13–18) and Stenbergen (blocks 19–21) pits fall well within each other's cone of 95 percent confidence (Table 2), indicating with much greater than 95 percent confidence that they could be drawn from one population, and

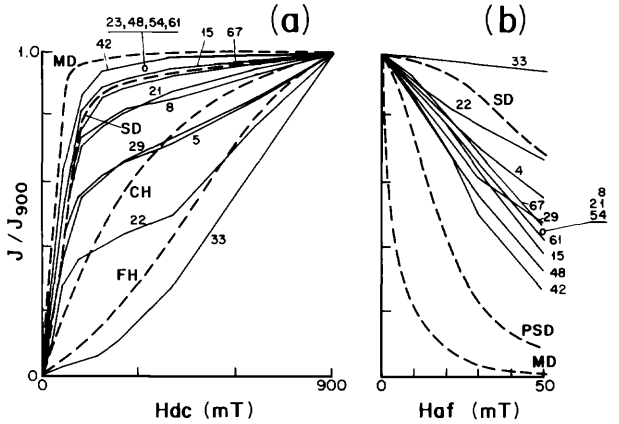


FIG. 7. Saturation isothermal remanent magnetization curves for specimens from the Central Missouri Mississippi Valley-type ore district. (a). Acquisition of saturation isothermal remanence. (b). Alternating field demagnetization of saturation isothermal remanence. J/J_{900} is the ratio of the measured remanence intensity to the saturation isothermal remanence intensity at 900 mT, and Hdc and Haf are the magnetizing and demagnetizing applied fields, respectively, in milliTeslas. Underlined numbers indicate additional curves falling between the two curves adjacent to the arrow. The type curves are for single domain (SD), pseudosingle domain (PSD), and multidomain (MD) magnetite and for fine-grained (FH) and coarse-grained (CH) hematite (from Dunlop, 1971, 1972, 1973, 1981).

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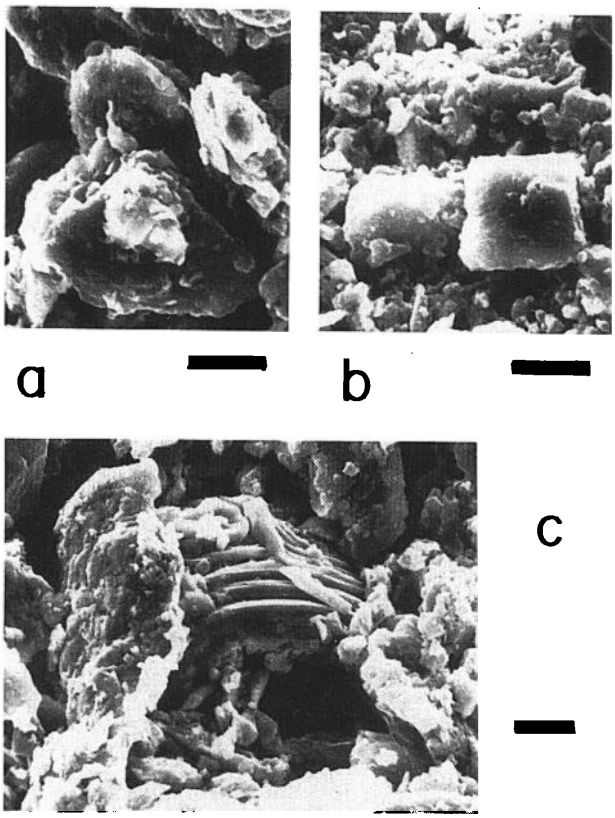


FIG. 8. Scanning electron microphotographs with 2-μm scale bars of magnetite crystals extracted from (a) and (b) A remanence component sites, showing typical blocky crystals, and (c) a B remanence component site showing an atypical replaced fossil.

therefore, that they can be combined into one population (McFadden and Lowes, 1981). Similarly the Goller-Stenbergen and Blackman (blocks 1–12) barite magnetizations are statistically the same, allowing all 21 directions to be combined into one population (Table 2).

The Jefferson City Dolomite in the highway 63 section also retains the A component with the lowermost seven blocks (50–56) forming a coherent cluster (Table 1, Fig. 6b). The somewhat divergent directions for the uppermost two blocks (57 and 58) are thought to be caused by some later contamination of the A magnetization, similar to the cause(s) of the B magnetization. The alternating field and thermal demagnetization characteristics (Fig. 7, block 54) for the seven blocks are essentially the same as those described for the barite zones except that their natural remanent magnetism intensities are distinctly weaker. SEM-EDA of the magnetic extract from the dolomite shows blocky crystals with iron peaks only that are typical of magnetite. Again the cones of 95 percent confidence for this dolomite section and for the barite veins encircle each others' mean magnetization direction (Table 2, Fig. 6c), indicating that they acquired their respective magnetizations at the same time.

The breccia blocks of the collapse structure in the highway 63 section should have randomly directed magnetizations if they retain a primary Early Ordovician remanence. When the eight block vectors (59, 61–67) are combined, they prove to be nonrandomly directed at the 95 percent confidence level using the test of Watson (1956) since the calculated value for vector resultant of 6.99 is very much greater than the significance point value of 4.48. This is a negative conglomerate test (Graham, 1949) which shows that the breccia blocks were remagnetized after collapse. The collapse structure is thought from stratigraphy to have formed after the mid-Devonian (J. Palmer, pers. commun.) and it is uncertain how far the blocks have dropped. Further, six of the blocks (61–65 and 67) clearly retain the A magnetization direction. The alternating field and thermal demagnetization characteristics (Fig. 2, block 67) and saturation isothermal remanence characteristics (Fig. 7, blocks 61 and 67) of the breccia blocks are very similar to those of the dolomite, indicating as before that single domain to pseudosingle domain magnetite is the main remanence carrier. The magnetite mineral extract from a breccia block that was examined by SEM-EDA, just as for the other A component samples, contains only blocky crystals (Fig. 8b) with peaks diagnostic of magnetite.

When the mean directions for the remanence of the Jefferson City Dolomite and the breccia fragments in the highway 63 section are compared, it is seen that the cone of 95 percent confidence of one circumscribes the mean of the other (Table 2). As before, this indicates that both are recording the same age of remanence acquisition and that both populations can be combined to form one population of magnetization directions. One block (23) from near the base of the silicified Gasconade Dolomite section on highway 50 gives the A component direction also, and so it is included in the population (Table 2).

A final test is to compare the A component mean directions for the barite zones with their Jefferson City Dolomite host rocks from the highway 63 section. As can be seen, the cone of 95 percent confidence for one encircles the mean of the other (Table 2, Fig. 6c), and the comparison of means test of McFadden and Lowes (1981) indicates that the two populations can be drawn from one population with >99.99 percent confidence. This shows that both the barite and host rocks acquired their magnetization at the same time and can be combined to form one population of A magnetization directions. The possibility cannot be ignored that the barite and the magnetite-borne remanence in the barite zone are unrelated. Specific gravity determinations show that the barite zone specimens contain about 13 percent barite on average,

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TABLE 2. Group Mean Remanence Data

Group	N	Dec	Inc	k	α_{95}
A component					
A Blackman barite zones (1–12)	12	148.1	19.8	30	8.1
B Goller barite zones (13–19)	6	154.2	14.6	18	16.2
C Stenbergen barite zones (19–21)	3	156.0	10.0	253	7.8
D Goller + Stenbergen (entries B + C)	9	154.2	13.0	27	10.0
E All barite zones (entries A + B + C)	21	151.1	16.9	28	6.2
F Highway 63 dolomite (50–56)	7	157.7	20.9	33	10.6
G Highway 63 breccia (61–65, 67)	6	154.0	16.4	19	15.8
H All highway 63 host rocks (entries F + G)	13	156.0	18.9	26	8.3
I All *A* component (entries E + H + 23)	35	153.5	17.6	27	4.8
B component					
J Widel barite zones (27–37)	11	25.5	72.9	16	11.7
K Manafee quarry (38, 40, 42, 43, 47–49)	7	80.8	84.4	12	18.1
L Highway 50 section (22, 24, 25)	3	50.3	76.0	7	51.1
M All *B* component entries (J + K + L)	21	36.8	77.8	13	9.2

Notes: Headings as in Table 1; parenthetical numbers for each group refer to block numbers in Table 1

ranging up to nearly 100 percent. If the barite were simply diluting the ambient magnetite content of the host rocks, then the latter should have a more intense remanence. However, the median remanence intensity after 20 mT alternating field demagnetization of 78 barite specimens is, in fact, 56 percent more intense than that of 45 Jefferson City Dolomite specimens. This indicates that addition of the A component coincided with addition of the barite, and therefore, that the barite's remanence is primary. The unlikely alternative is that there were two postbrecciation precipitation events—one of barite and one of magnetite—that were related to enhanced permeability or porosity in the breccia but were widely separated in time (>30 m.y.). The single crystal tests were run to test for this alternative.

The white barite crystal has a strong natural remanence of 8.1×10^{-4} A/m, typical of the 100 percent barite specimens, that the alternating field demagnetizes to give a linear decay vector direction. Conversely, the blue barite crystal has a very weak natural remanence of 2×10^{-5} A/m. Both barite crystals give saturation isothermal remanence acquisition characteristics typical of magnetite and the decay of this remanence on alternating field demagnetization indicates pseudosingle domain behavior (Fig. 9). This is strong evidence that the barite crystals were precipitated with magnetite inclusions as one event and not as two separate events. The galena crystal gave an intense natural remanence of 7.2×10^{-3} A/m or about one hundred times an average dolomite specimen from this collection. Thus trace amounts of galena could be a major contributor to their remanence since Coveney et al. (1987) have shown the widespread occurrence of trace amounts of galena in the carbonates of the midcontinent. An artificial specimen with about 2 percent galena in plaster of paris shows saturation isothermal remanence acquisition and alternating field demagnetization characteristics that are typical of pseudosingle domain magnetite. The specimen with sphalerite crystals has a relatively strong natural remanence intensity of 4.5×10^{-4} A/m, and so trace amounts of this mineral could also contribute to the remanence. Its isothermal remanence acquisition and alternating field demagnetization of saturation isothermal remanence characteristics are indicative of pseudosingle domain hematite with minor pseudosingle domain magnetite (Fig. 9). These single crystal tests show that Mississippi Valley-type ore minerals carry a magnetization with characteristics that match those of the A component specimens, making the argument for separate Mississippi

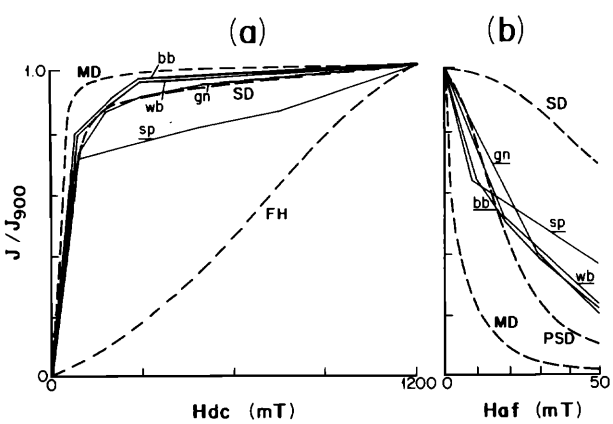


FIG. 9. Saturation isothermal remanent magnetization curves for single crystals of white barite (wb), blue barite (bb), and sphalerite (sp) and of galena powder from a single crystal dispersed in plaster of paris (gn). Other conventions as in Figure 7.

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Valley-type ore mineralization and magnetization highly improbable.

B Component

All of the baritic zone blocks from the Widel property (27–37) retain the B magnetization direction (Table 1, Fig. 6). On alternating field step demagnetization, the remanence intensity of these samples is rarely reduced to less than 10 percent of the natural remanence intensity by 140 mT (Fig. 3, block 22), suggesting the presence of some hematite. This is confirmed in a few specimens on thermal step demagnetization where the Néel temperatures can exceed 650°C (Fig. 5, block 32). Most of the specimens, however, show significant intensity decay in the magnetite blocking temperature range and direction that is characteristic of a Cenozoic viscous overprint residing in pseudosingle domain to multidomain magnetite (Jackson, 1990). There is also a very minor offset of the vector close to the origin that indicates the presence of hematite with a remanence intensity that is close to the noise level of the magnetometer. The saturation isothermal remanence curves provide further evidence, with the acquisition and decay curves being characteristic of fine-grained hematite in one case (Fig. 7, block 33) and of mixed magnetite and hematite in the other case (Fig. 7, block 29). The magnetic extract from high-grade barite ore contains blocky crystals with iron peaks only except for one tubular structure (Fig. 8c) with iron peaks only that looks like a fossil that has been partially replaced by magnetite. Thomsen and Voren (1984) have reported pyrite framboids and pyritized tubes and burrow fillings from a few meters below surface in Pleistocene glaciomarine sediments. Our fossil could be a short segment of a pyritized tube that has been replaced by magnetite as has been recently observed for framboids in Paleozoic carbonates by Suk et al. (1990). The fact that the Widel samples came from surface exposures suggests that their hematite remanence is a weathering product and that their magnetite remanence reflects replacement during the Quaternary or resetting of an earlier magnetite by lightning strikes (Tarling, 1983) during the Quaternary. This combination would explain the dispersed population of mean remanence directions about the present earth's magnetic field as well as the rock magnetic characteristics.

The highway 50 section of silicified Gasconade Dolomite extends only a few meters below obvious weathering. The uppermost block (26) was one of only two in the entire collection that did not provide a coherent remanence. The next two blocks downward (24 and 25), presumably the next most susceptible to weathering and/or lightning, give the B magnetization direction, as does the lowermost block (22) (Fig. 6e). The intervening block (23) retains the A direction and was added to the A population. Again alternating field and thermal step demagnetization of the B component specimens suggest the presence of both magnetite and hematite (Fig. 3, block 22). The saturation isothermal remanence curves (Fig. 7, block 22), especially the acquisition curve, is diagnostic of the presence of significant amounts of both magnetite and hematite.

The specimens from the Manafee quarry section of the Mississippian Burlington Limestone have the weakest natural remanence intensities in the collection, being mostly in the low to middle 10^{-5} A/m range. They also have two types of remanence directions. The first type are those with the B component (blocks 38, 40, 42, 43, and 47–49; Fig. 6e). They show a relatively rapid intensity decay down to $<5 \times 10^{-6}$ A/m range on alternating field demagnetization (Fig. 3, block 42), and they lose directional consistency above 500°C on thermal demagnetization (Fig. 5, block 49). They saturate rapidly on being magnetized and their saturation isothermal remanence is demagnetized rapidly in the pseudosingle domain to multidomain range (Fig. 7, blocks 42 and 48). These properties indicate that the weak limestone remanence is carried by relatively large magnetite grains with relatively low coercivity and blocking temperature spectra when compared to the rest of this collection. In the magnetic extract from the Burlington Limestone, the only iron-bearing crystals are relatively large and irregularly shaped magnetites. These features suggest that the limestone specimens are the most likely ones in this collection to have been remagnetized. The second type of remanence directions (blocks 39, 41, and 44–46) has shallow inclinations to the northwest. Except for direction, it has the same magnetic properties as the B component group. The most probable explanation for their deviant directions—proffered without clear evidence—is that they record a hybrid magnetization caused by incomplete remagnetization. This could be caused, for example, by a combination of a Quaternary B component and a primary Mississippian component which would be directed to the northwest with an upward inclination of about 20°.

In the highway 63 road cut of the Jefferson City Formation, the uppermost two blocks of dolomite (57 and 58), which are the most likely to have been altered by weathering, and two breccia blocks (59 and 66) give magnetization directions that deviate from the A cluster. These specimens show evidence of mixed remanence components on demagnetization by the failure of the characteristic magnetization vector to decay through the origin of an orthogonal plot (Fig. 5, block 66).

When the B component directions are averaged for the Widel pit, Manafee quarry, and highway 50 section, respectively, they give means that fall within

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the cones of 95 percent confidence of the other means (Table 2). This agreement indicates that they acquired their B magnetizations at the same time and that they can be combined into one population (Table 2).

Discussion

The barite zones in the Blackman, Goller, and Stenbergen pits, the Lower Ordovician Jefferson City Dolomite, one block of silicified Gasconade Dolomite, and the large fragments in the collapse breccia of post-Middle Devonian age in the Jefferson City Dolomite, all acquired their A magnetization at the same time. The breccia block test shows that this A magnetization is secondary in the breccia and dolomites, and therefore Middle Devonian or younger. The pole position for the A component is located at 121.0°E, 36.6°N (dp = 2.6°, dm = 4.9°). Its oval of 95 percent confidence falls on the Late Pennsylvanian to early Early Permian portion of the apparent polar wander path for the North American craton of Van der Voo (1990) to define the age of the A component (Fig. 10). This age also falls within the Kiaman superchron from Early Pennsylvanian to Late Permian during which the earth's magnetic field was reversed for about 96 percent of the time (Irving and Pullaiah, 1976). Our results agree with this limitation in that all of our A component specimens are reversed. Recalling the increased remanence intensity found in the baritic specimens and the magnetic properties of the barite single crystals, it is virtually certain that the barite remanence is synchronous with mineralization, and therefore, that both the barite and A remanence component were emplaced in Late Pennsylvanian to early Early Permian time. At the same time precipitates from the transient brines must have produced the A component in the breccia blocks, and so it is likely that the blocks were cemented together to form a solid rock at that time. Finally, the magnetization in the Ordovician host rock was similarly acquired at this time by precipitation of magnetite from brine either during dolomitization of a preexisting limestone or during the infilling of a preexisting porous dolomite.

Baritic samples from the Widel property, the Burlington Limestone in the Manafee quarry, and most of the blocks of silicified Lower Ordovician Gasconade Dolomite in the highway 50 road cut, all carry the B magnetization component. It is carried by both magnetite and hematite in the Widel samples where there is SEM evidence of magnetite replacement of a fossil. The B component in the highway 50 dolomites is clearly related to modern weathering. In the Burlington Limestone, the B component is carried in magnetite domains with relatively low blocking temperature and coercivity spectra that could be recording a relatively stable viscous or isothermal remanent magnetization of Quaternary age. The paleopole for

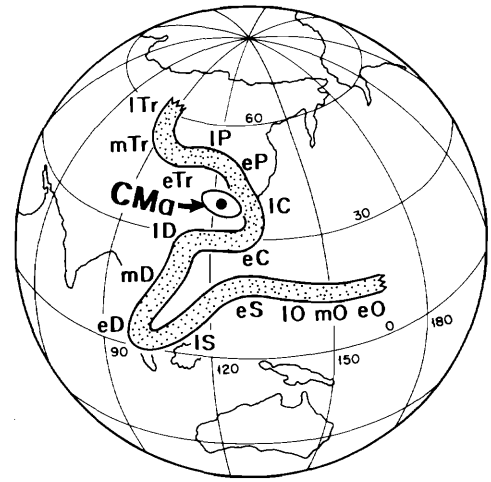


FIG. 10. Apparent polar wander path for the North American craton from Van der Voo (1990) for the Early Ordovician (eO) to Late Triassic (lTr) with the Central Missouri A component pole (CMa) surrounded by its oval of 95 percent confidence.

the component is in northern Quebec at 68.4°N, 55.4°N (dp = 16.2°, dm = 17.3°). This paleopole is well off any part of the Mississippian to present apparent polar wander path (Irving and Irving, 1982; Van der Voo, 1990) and therefore must be a hybrid magnetization. Several factors indicate that the earlier A component in the magnetite has not been entirely overprinted by the later Quaternary B magnetization. First, the B direction falls on a great circle between the A direction and the present earth's magnetic field time-averaged dipole direction of declination = 0°, inclination = 58° for central Missouri (Fig. 6c). Second, the rocks carrying the B magnetization must have been exposed to the same brine solutions that produced the A magnetization in magnetite, as was witnessed in the lower part of the highway 50 Gasconade section. Third, the rocks from the Widel property and highway 50 section that carry significant modern hematite remanence in addition to magnetite remanence give a mean direction that is closer to the present dipole direction than do the rocks from the Manafee quarry that carry only magnetite remanence. Fourth, the A pole falls on the polar wander path and no other known component in combination with B could produce A to give this pole.

The Late Pennsylvanian-early Early Permian or Kiaman age for the barite deposition and for the other A remagnetization events in central Missouri is important from the broader perspective of the genesis of Mississippi Valley-type ore deposits in the surrounding districts. Despite the fact that the lead-zinc deposits in the Viburnum Trend-Old Lead belt district of southeastern Missouri occur in Cambrian rocks and those in the Northern Arkansas district occur in Ordovician and Mississippian rocks, the ores and host

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rocks of both districts also give Kiaman magnetization directions (Beales et al., 1974; Wu and Beales, 1981; Wisniowiecki et al., 1983; Dunn and Elmore, 1985; Pan et al., 1990; Fig. 1). Ore samples from the Tri-State district, collected from old surface workings, all gave modern remanence directions from weathering (Pan et al., 1990). However, the similarity in style of mineralization, the mineral assemblage such as the abundance of chert not found in other districts, and the presence of numerous small deposits of similar ilk between the Northern Arkansas and Tri-State districts (McKnight, 1935; Brockie et al., 1968; Leach et al., 1975) provide a reasonable assurance that the two districts simply represent two major concentrations within a single larger district, and that the northern Arkansas paleomagnetic results likely apply equally to the Tri-State district. Thus, three major Mississippi Valley districts (Central Missouri, Southeastern Missouri, Tri-State-Northern Arkansas; Fig. 1), spread over a 240,000-km^2 region of the midcontinental United States appear to have been mineralized at essentially the same time. This is exactly the result that one would expect following the tectonic deformation and uplift model for Mississippi Valley-type ore genesis proposed by Leach (1979), Leach et al. (1984), and Leach and Rowan (1986). Based on evidence of a northward decrease in fluid inclusion homogenization temperatures, these authors theorized that hot brines migrated out of the Arkoma basin upon uplift during the Pennsylvanian to Permian Ouachita orogeny as the Llanorian and North American plates collided. The lead-zinc-barite deposits were subsequently precipitated from the northward-migrating brines throughout the foreland carbonate platform. The possibility raised by Leach (1980) that the lead-zinc and barite deposits might represent two mineralizing events can be accommodated if both events occurred within the Kiaman superchron from Middle Pennsylvanian to the end of the Early Permian.

Garven and Freeze (1984a and b) provided a mathematical basis for gravity-driven fluid migration out of a basin using a hydraulic head in an adjacent highland. In the Missouri region of the midcontinent, because the adjacent Ouachita Mountains have been present since the Pennsylvanian, this model implies Mississippi Valley-type ore formation on a continuous basis since the Pennsylvanian. Thus the model is in conflict with the paleomagnetic results, unless mineralization is attributed to only the first flushing of brine from the Arkoma basin on initial uplift as Bethke et al. (1988) calculated. Oliver (1986) extended the gravity-driven fluid flow model to the Appalachians where the ores and host rocks of the East Tennessee district also give a Kiaman magnetization that is coeval with the Alleghanian orogeny (Bachtadse et al., 1987). Further many other Paleozoic carbonate and red-bed units in the Ozark and Appalachian regions are now known to carry a Kiaman remagnetization, reflecting the pervasive nature of the flowing intraformational fluids at this time (e.g., Scott, 1979; Scotese et al., 1982; McCabe et al., 1983; Roy and Morris, 1983; Kent and Opdyke, 1985; Irving and Strong, 1985; Miller and Kent, 1986; McCabe et al., 1989).

Other models for the genesis of Mississippi Valley-type ore deposits do not give a particularly good fit to the paleomagnetic data from the Central Missouri, Southeastern Missouri, and Northern Arkansas-Tri-State districts. The "basin compaction" model of Jackson and Beales (1967) relies on compaction of basinal sediment to drive the brines to the basin periphery continually. In these districts this should lead to a spectrum of Paleozoic remagnetization ages as the surrounding basins were filled in. The episodic basin dewatering by compaction model of Cathles and Smith (1983) suggests that rapid basin infilling and sediment compaction results in overpressures which periodically expelled brines laterally out of the basin shales and into coeval carbonates in the platform with maximum expulsion occurring about 40 m.y. after initial burial of the basinal sediments. Thus this model also predicts a spectrum of Paleozoic ages for the Mississippi Valley-type deposits in the midcontinental region rather than the single, relatively short Kiaman interval that is found.

Conclusions

Most barite samples from the Central Missouri district carry a primary magnetization of Late Pennsylvanian to Early Permian age from within the reversed Kiaman superchron. Their host rocks in the Ordovician Jefferson City Dolomite were remagnetized at the same time as shown by a negative conglomerate test in a collapse breccia in the Jefferson City Dolomite. The magnetite remanence of surficial samples of barite has been overprinted during the Quaternary by hematization from weathering, by Recent replacement, or by lightning-induced viscous remanence.

When added to earlier paleomagnetic studies that report Kiaman magnetizations in the lead-zinc ores and host rocks of the Southeastern Missouri and Northern Arkansas districts, this study supports the tectonic uplift model described by Leach and Rowan (1986) for the genesis of these lead-zinc and barite deposits. The results do not support the basin compaction models of Jackson and Beales (1967) or of Cathles and Smith (1983).

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